TTEdit: Cross-Modal Fusion with Diffusion Models for Detail-Aware Fashion Editing

Feng Zhang[1,3,5], Junliang Tan[1,2], Zhenming Chen[1], Hao Feng[1], Biao Guo[1], Junyan Chen[1], Yao Lu[1], and Ming Jiang[1,3,4](✉)

[1] School of Computer Science and Information Security, Guilin University of Electronic Technology, Guilin 541004, China
zhangfeng@guet.edu.cn
[2] Guangxi Academy of Artificial Intelligence, Nanning 530201, China
[3] Guangxi Metaverse Application Engineering Center, Guangxi Institute of Digital Technology, Nanning 530000, China
mjiang@guet.edu.cn
[4] School of Information Engineering, Guangxi Polytechnic of Construction, Nanning 530003, China
[5] Guangxi Guanyang Digital Technology, Nanning 530000, China

Abstract. Recent advances in text-driven image editing using visual language models demonstrate considerable potential. However, purely text-based methods face persistent challenges in fashion image editing, where a semantic gap hinders the accurate expression of complex garment details and the generation of visuals faithful to textual descriptions. While FashionTex pioneered the use of texture images as auxiliary inputs with GANs for local editing, inherent limitations of GANs constrain generative quality. To address these issues, we propose **TTEdit**, a novel diffusion model-based framework for fashion image editing. TTEdit innovatively integrates dual-modality information from text descriptions and texture images to achieve precise, localized garment edits. Our approach consists of two core components: (1) **MCNet**, a lightweight and accurate localization module built upon mainstream segmentation models to identify editing regions, and (2) the **Texture-Adapter**, which injects fabric texture features into the diffusion model via a cross-modal attention mechanism. Extensive experiments demonstrate that TTEdit significantly outperforms existing methods, achieving state-of-the-art results with an FID of 11.55, SSIM of 0.871, and CLIP-I of 0.83. These metrics surpass the previous best method, Insert Anything (FID 11.98, SSIM 0.848, CLIP-I 0.81), confirming our framework's superior generative quality, fidelity, and controllability.

Keywords: Fashion Image Editing · Multimodality · Diffusion Model

1 Introduction

Image editing is a key technology for digital content creation. It involves semantic modification, quality enhancement, and content regeneration of existing images

J. Lokoč et al. (Eds.): MMM 2026, LNCS 16413, pp. 290–303, 2026.
https://doi.org/10.1007/978-981-95-6957-1_21

[1]. The scope ranges from color correction to semantic reconstruction, supporting applications such as beautification, style transfer, and content completion. At the intersection of computer vision and graphics, deep learning methods provide users with creative freedom and precise control [2].

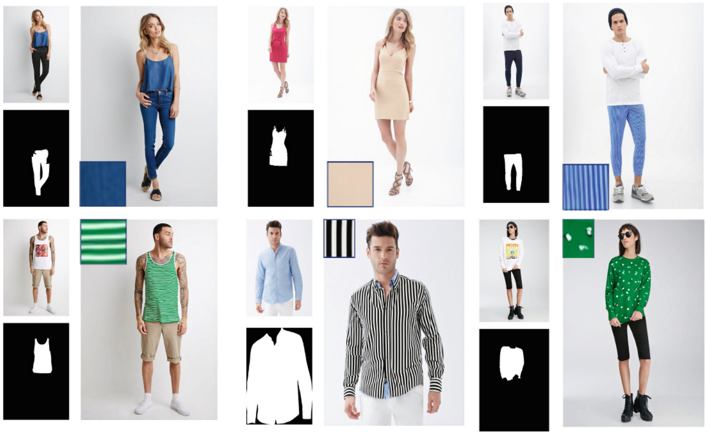

Fig. 1. TTEdit. Users can edit clothing regions by painting with a reference texture. Our method automatically adapts the reference image and seamlessly integrates it into the source image, enabling high-quality editing results.

Generative models have transformed image editing. Early methods based on GANs [3] achieved initial success but suffered from unstable training, mode collapse, and artifacts. Diffusion models [4], in contrast, generate high-fidelity images through a progressive denoising process, offering better stability.

Vision–language models further enabled text-driven image editing. By leveraging cross-modal pretraining such as CLIP [5], image editing can be guided by natural language. This paradigm reduces the technical barrier and allows non-experts to engage in creative tasks. Despite these advantages [6–8], text-driven methods remain limited in fashion editing. They struggle with precise region localization and often misinterpret fine-grained semantics, leading to inaccurate visual outputs.

FashionTex [9] addresses semantic limitations with reference textures, improving semantic–visual alignment. However, as a GAN-based method, it still suffers from mode collapse and low image quality. More critically, it lacks the ability for precise local editing without disturbing unrelated regions.

We propose TTEdit to overcome these challenges. The MCNet module, a lightweight dual-branch image–text encoder, collaborates with a segmentation model to locate target regions accurately. In the rendering stage, the Texture-Adapter employs multi-scale feature injection, dynamically aligning texture features with UNet layers. This ensures consistency from local details to global structure. Experiments demonstrate that TTEdit produces superior image quality, realistic textures, and smooth editing boundaries. Our main contributions are:

- We propose the TTEdit editing framework, which combines text with texture images and diffusion models, effectively solving the problems of insufficient text description representation and inability to accurately locate the editing region.
- In order to improve the editing accuracy, we designed MCNet, which can assist the segmentation model to locate the garment regions that need to be edited and provide precise guidance for subsequent image modification.
- To effectively utilize texture details as conditional guidance for diffusion-based generation, we introduce a Texture-Adapter module. Furthermore, we improve the formulation of texture loss to produce more realistic and visually consistent results.
- Experimental results show that our method outperforms the existing methods in a number of metrics, including image fidelity, local consistency, and user judgments, and achieves a more realistic and detailed fashion image editing effect.

2 Related Work

Text-Driven Image Editing. With the rapid progress of text-driven image generation, models such as DALL-E 2 [10] and Stable Diffusion [11] enable image creation directly from natural language. InstructPix2Pix [13] further extends this paradigm to instruction-based editing, allowing images to be modified according to natural language guidance. In fashion editing, Texfit [14] explores modifying clothing styles, patterns, and colors through textual input, supporting flexible personalization. To improve semantic alignment, methods such as GLIDE [15], Textual Inversion [16], and T2I-Adapter [17] strengthen the mapping between language and images. While these approaches are effective in generating diverse content, they still struggle with fine-grained control, particularly for complex textures and material details. To address this, we introduce reference texture images as additional conditions, enhancing texture representation and yielding more realistic editing results.

Reference-Based Fashion Editing. A large body of work focuses on transferring target clothing onto user images. TryOnGAN [18] applies layered interpolation to improve pose consistency, while PASTA-GAN++ [19] leverages spatial guidance and Texture-Adapter modules for more realistic try-on results. Fashion-Tex [9] combines texture references with text prompts to enhance visual fidelity and detail control. Despite these advances, most approaches rely on GANs, which face limitations in fine-grained texture modeling and handling complex geometric deformations. As a result, synthesized results often exhibit misaligned patterns, stretched fabrics, or unnatural folds [20,21], reducing overall visual consistency. In this work, we leverage reference texture information within diffusion models, achieving more controllable, high-quality, and natural fashion image generation.

3 Method

To achieve personalized garment editing, we propose a method that combines prompt-based conditional control with reference texture images. In Fig. 2, this is the architecture diagram of our model, given a full-body image I, our approach employs MCNet to use textual prompts to assist the general segmentation models in accurately selecting the mask M of the target garment region corresponding to the description of the user. Subsequently, the selected region is rendered using the reference texture image in conjunction with the garment prompt. The implementation details are presented below.

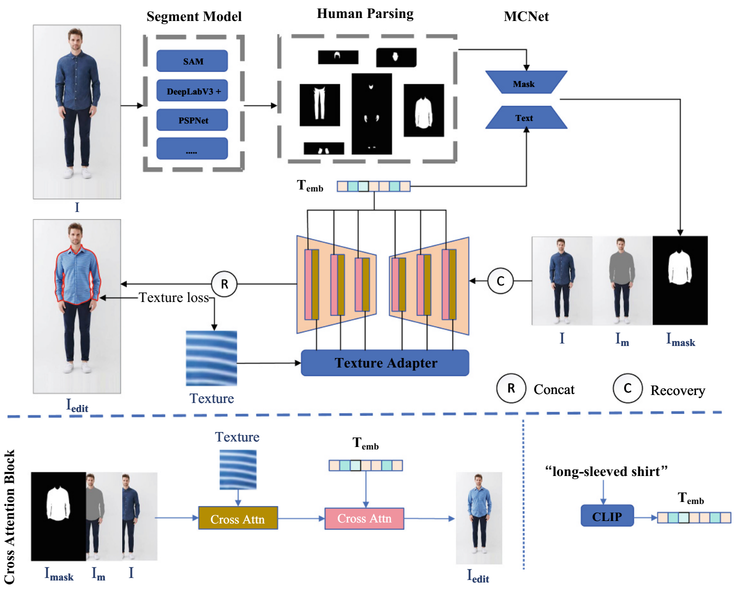

Fig. 2. Overview of our pipeline.

3.1 Region Extraction for Editing

Given that mainstream segmentation models (e.g., SAM [22], DeepLabV3+ [23], and PSPNet [24]) have demonstrated strong performance in human parsing, we use these models to parse the input image I, obtaining a detailed segmentation of the human body and the corresponding segmentation results.

To further refine the selection of the target garment mask, we design MCNet, a network that assists the segmentation model in mask selection. MCNet consists of two types of encoder: a mask image encoder E_{mask} and a text encoder T_{emb}. These encoders extract features from two different modalities—visual mask images and textual garment descriptions.

During training, we introduce a multi-modal semantic alignment loss, which aims to align the semantic space of the garment prompts with that of the corresponding mask features. This encourages the model to better understand the correspondence between natural language descriptions and visual regions. The alignment loss function is defined as follows:

$$\mathcal{L}_{\text{alignment}} = -\sum_{i=1}^{N} \log \left(\frac{\exp\left(\text{sim}\left(f_I\left(\text{aug}_I(M_i^+)\right), f_T(T_i^+)\right)/\tau\right)}{\sum_{j=1}^{N} \exp\left(\text{sim}\left(f_I\left(\text{aug}_I(M_i^+)\right), f_T(T_j)\right)/\tau\right)} \right) \tag{1}$$

where $f_I(\cdot)$ denotes the image encoder used to extract features from visual inputs, $\text{aug}_I(\cdot)$ represents the data augmentation function applied to the mask regions, M^+ is the positive sample feature extracted from the target mask region, T^+ is the positive sample feature corresponding to the associated garment text prompt, T_j refers to the j-th text feature among all candidate textual descriptions, and τ is the temperature parameter controlling the smoothness of the similarity distribution.

We encourage the similarity between the mask region and its corresponding ground-truth text to be maximized, while minimizing its similarity with other non-matching textual descriptions. In Fig. 3, the image encoder processes the input conditions, followed by multi-scale processing with downsampling operations between different scales, enabling high-quality texture extraction and generation.

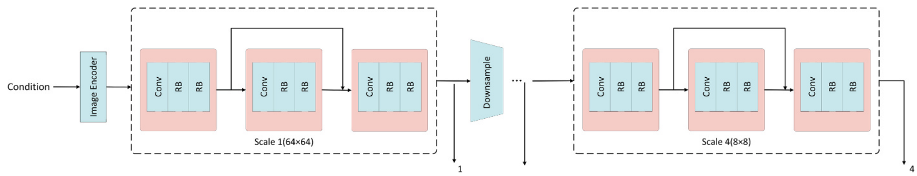

Fig. 3. Texture-Adapter Network Architecture.

During inference, we utilize the trained MCNet to evaluate the segmented mask regions produced by the parsing model. MCNet automatically selects the target mask region M_T that is semantically most relevant to the input prompt. This region represents the garment area to be edited, and is subsequently used for texture generation or image editing tasks. This enables more accurate garment manipulation and texture replacement. The selection process can be formulated as follows:

$$M^* = \arg\max_{M_i} \text{sim}\left(\phi_v(I \odot M_i), \phi_t(T)\right) \tag{2}$$

where $\phi_v(\cdot)$ denotes the image encoder used to extract features from visual inputs, $I \odot M_i$ represents the masked image obtained by applying mask M_i to the original image I, $\text{sim}(\cdot, \cdot)$ is the similarity function that measures the closeness between two feature vectors, M^* is the selected target mask region most semantically relevant to the input text prompt, $\phi_t(\cdot)$ denotes the text encoder used to extract features from the textual description T.

3.2 Content Generation

Due to the insufficient control over the generated content by text description alone, we leverage reference texture images to assist generation. The texture prompts mainly provide the target image with surface material, detail distribution, and style attributes, characterized by strong high-frequency information and strict spatial consistency requirements. To effectively extract semantic information from the texture images, we employ a Texture-Adapter to extract multi-level features. These texture semantic-rich features are injected into the cross-attention modules of the U-Net. Specifically, for each query feature Z in the U-Net module, we use the texture feature c_τ as a key and value to perform the cross-attention computation:

$$Z_{\text{texture}} = Softmax\left(\frac{QK^T}{\sqrt{d}}\right)V \tag{3}$$

$$Q = ZW_q, K = c_\tau W_k, V = c_\tau W_v \tag{4}$$

where W_q, W_k, W_v are the learnable weights trained during the process, c_τ denotes the projected texture prompt sequence.

This mechanism endows the diffusion process with texture-aware capability, enabling precise control over material and detail distribution in image generation.

To incorporate the semantic guidance of clothing-related textual prompts, we use a pre-trained CLIP text encoder to encode the input prompt and obtain a global text embedding. This embedding is then projected through a trainable projection network to generate a sequence of text embeddings c_{text}, which are subsequently injected into the cross-attention modules of the U-Net:

$$Z_{\text{text}} = \text{Softmax}\left(\frac{Q_{text}K_{text}^{\text{T}}}{\sqrt{d}}\right)V_{\text{text}} \tag{5}$$

$$Q_{text} = ZW_q^{\text{text}}, K_{text} = c_{text}W_k^{\text{text}}, V_{text} = c_{text}W_v^{\text{text}} \tag{6}$$

where $W_q^{\text{text}}, W_k^{\text{text}}, W_v^{\text{text}}$ are learnable parameters during training, and c_{text} is the projected text embedding sequence. The final fused output is defined as:

$$Z_{\text{new}} = Z + Z_{\text{texture}} + Z_{\text{text}} \tag{7}$$

During the training stage, we jointly train the Texture-Adapter and the U-Net network. The training dataset consists of paired textual prompts, reference texture images, and target images. The training objective is to minimize the diffusion reconstruction error using the standard denoising loss function:

$$\mathcal{L}_{\text{simple}} = \mathbb{E}_{x_0,\epsilon,c_\tau,c_{\text{text}},t} \left\| \epsilon - \epsilon_\theta(x_t, t, c_\tau, c_{\text{text}}) \right\|^2 \tag{8}$$

To preserve structural consistency while effectively transferring style characteristics (e.g., texture, detail, material) from the reference texture image to the target image, and simultaneously ensuring semantic consistency with the textual prompt, we employ the following style transfer loss:

$$\mathcal{L}_{\text{style}} = \sum_{l \in \mathcal{C}} \lambda_l \cdot \|G_l(I_g) - G_l(I_s)\|_F^2 \tag{9}$$

where λ_l is the weight hyperparameter for each layer to balance contributions across layers, and $\|\cdot\|_F$ denotes the Frobenius norm, which measures the difference between two Gram matrices. $G_l(I_s)$ and $G_l(I_g)$ represent the style features of the reference and generated images at the l-th layer, respectively.

To achieve high-quality and personalized clothing texture editing, we jointly optimize multiple objectives during training. Specifically, we adopt the standard diffusion reconstruction loss $\mathcal{L}_{\text{simple}}$ to guide the overall image generation. Meanwhile, to ensure consistency in both texture appearance and semantic alignment, we define a texture loss $\mathcal{L}_{\text{texture}}$ that combines style transfer loss $\mathcal{L}_{\text{style}}$ and semantic alignment loss $\mathcal{L}_{\text{alignment}}$:

$$\mathcal{L}_{\text{texture}} = \beta_1 \cdot \mathcal{L}_{\text{style}} + \beta_2 \cdot \mathcal{L}_{\text{alignment}} \tag{10}$$

The total training objective is then defined as:

$$\mathcal{L}_{\text{total}} = \alpha \cdot \mathcal{L}_{\text{simple}} + \mathcal{L}_{\text{texture}} \tag{11}$$

where α is the weight coefficient for the loss of diffusion reconstruction, while β_1 and β_2 control the importance of style fidelity and semantic alignment in the texture region.

In our experiments, we set $\alpha = 1$, $\beta_1 = 25$, and $\beta_2 = 20$ to ensure high-quality texture transfer and precise semantic control [25].

During the inference stage, given a texture image and a text prompt as input, the model is capable of generating image results with consistent material style and detailed texture.

4 Experiments

4.1 Experimental Settings

Dataset. In this study, the Deepfashion-MultiModal dataset [26] is used for the training and evaluation of the methods. This dataset is a multi-modal fashion dataset containing 12,701 high-quality full-body human images, each labeled with (1) detailed text descriptions, (2) 24 categories of human parsing labels, and (3) human keypoint detection data. To ensure the reliability of the experiments,

Fig. 4. Qualitative comparison with other approaches. Our method can generate results that are semantically consistent with the input reference images in high perceptual quality.

we divided the dataset into a training set (11,260 images) and a test set (1,441 images) according to standard protocols. In addition, considering the specific needs of this study, we additionally integrated local texture samples from the FashionTex dataset as supplementary reference data to enhance the model's ability to perceive garment texture features.

Baselines. To comprehensively evaluate the effectiveness of our proposed framework, we select five representative diffusion-based editing methods as baselines, covering both text-driven and reference-driven paradigms. The comparison setting is carefully designed to ensure fairness and reproducibility.

InstructPix2Pix [13] and TexFit [14] are representative text-guided editing approaches based on diffusion models. InstructPix2Pix learns to follow natural language editing instructions for general image manipulation, while TexFit specifically targets fashion image editing with text-driven control. Since both methods rely primarily on textual conditions, we first employ BLIP [12] to generate descriptive captions from the reference texture images. These captions are then used as textual inputs to InstructPix2Pix and TexFit, ensuring that the models are provided with semantic instructions derived from the texture itself. This setting allows the text-driven models to generate outputs consistent with the intended clothing patterns and styles.

In contrast, Paint by Example [27], AnyDoor [28], and Insert Anything [29] adopt reference-driven paradigms. Paint by Example performs exemplar-based editing by conditioning the diffusion process on the appearance of a reference image, enabling direct transfer of texture details to the target region. AnyDoor extends this idea to object-level customization in a zero-shot manner, allowing seamless transfer of visual concepts across contexts without additional training. Insert Anything leverages in-context editing within a DiT backbone, supporting flexible insertion of visual content into a target scene guided by an exemplar. For these methods, we directly provide the reference texture image as the condition, which allows them to utilize their intrinsic exemplar-based capabilities for texture transfer and editing.

Implementation Details. All our experiments were run on a single NVIDIA Tesla V100 graphics card. To accelerate training and improve computational efficiency, we resized all images to a resolution of 512×256.

For MCNet training, we used the following parameter configuration: 50 epochs were trained, batch size was 8, the learning rate was set to 1×10^{-4}, and the optimization was performed using Adam Optimizer. We adopt a two-stage training strategy to optimize the classification network and the image generation model separately.

In the first stage, we fine-tune MCNet, a classification network built on the CLIP architecture. To better adapt to our specific dataset and task, both the CLIP text encoder and image encoder are fine-tuned during this stage. The training parameters are set as follows: a total of 150k fine-tuning steps, a batch size of 8, and a learning rate of 5×10^{-6} to maintain the semantic alignment of the pre-trained CLIP model. Due to memory constraints, we employ mixed-precision training and adopt a gradient accumulation strategy with 4 accumulation steps.

In the second stage, we fine-tune the image generation model based on Stable Diffusion v1.5. Specifically, both the U-Net module and the Texture-Adapter are jointly trained, while all other components remain frozen to preserve the pre-trained knowledge. The training configuration remains consistent with the first stage, including 150k fine-tuning steps, a batch size of 1, and a learning rate of 1×10^{-5}. Similarly, mixed-precision training and gradient accumulation with 4 accumulation steps are applied to enhance training efficiency.

In the inference stage, we set the generation parameters to 50 iterations and the classifier-free guidance coefficient to 7.5.

4.2 Comparative Results

To evaluate the effectiveness of our method, we compare it with representative texture transfer approaches, as shown in Fig. 4. The results demonstrate that our method achieves more stable performance in texture migration, accurately preserving the fine-grained details, material properties, and style consistency of the reference textures.

For text-driven editing methods, a common pipeline first applies BLIP [12] to the texture reference image to generate a textual description containing material, pattern, and color information. This description is then fed into frameworks such as InstructPix2Pix [13] and TexFit [14], whose performance, however, deteriorates with complex prompts, producing less accurate texture reproduction. Paint by Example [27] tends to lose fine texture details, leading to overly smoothed or faded appearances. AnyDoor [28] better maintains global style but suffers from distortions and pattern inconsistencies in complex regions. Insert Anything [29] focuses on local insertions yet often introduces texture blending artifacts or geometric mismatches. In contrast, our approach consistently generates results that are both visually realistic and semantically consistent.

For quantitative evaluation, Table 1 reports results on four widely used metrics: Fréchet Inception Distance (FID) for measuring realism, Structural Similarity Index (SSIM) for structural fidelity, Learned Perceptual Image Patch Similarity (LPIPS) for perceptual consistency, and CLIP-I for semantic alignment. In addition, we conducted a user study in which 30 human annotators were asked to evaluate 100 generated images based on realism and texture consistency.

Table 1. Quantitative comparison with diffusion-based reference-guided methods.

Methods	FID ↓	SSIM ↑	LPIPS ↓	CLIP-I ↑	User Study ↑
InstructPix2Pix [13]	13.88	0.769	0.141	0.61	53
TexFit [14]	12.49	0.829	0.129	0.68	62
Paint by Example [26]	16.34	0.788	0.138	0.72	60
AnyDoor [27]	12.77	0.833	0.125	0.74	73
Insert Anything [28]	11.98	0.848	**0.119**	0.81	71
TTEdit (Ours)	**11.55**	**0.871**	0.123	**0.83**	**80**

4.3 Ablation Studies

In order to deep analyze the core mechanism of our method and verify the effectiveness of each key component, we designed two sets of systematic ablation experiments to examine (1) the effect of prompt conditional control and (2) the effect of the texture loss function.

The Effect of Prompt Conditional Control. In Fig. 5, the comparison experiment clearly demonstrates the influence of cue word condition control on the generation effect. The cue word control can effectively help the model to understand the relationship with the mask region, in order to effectively generate the dress corresponding to the texture.

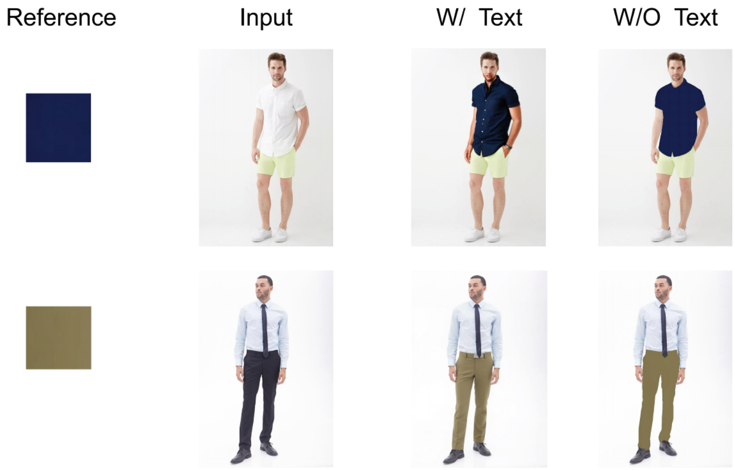

Fig. 5. Results of ablation experiments with the cue word conditional control module. Each row represents a set of dress style migration experiments, with the first column being the color cue for the reference image, the second column being the input character image, the third column being the results after the introduction of the cue word conditional control module (W/ Text), and the fourth column being the results without the introduction of the module (W/O Text).

The Effect of Texture Loss Functions. In Fig. 6, The introduction of the texture loss results in a significant performance improvement. As shown in Table 2, the adoption of the proposed loss results in notable enhancements across key metrics including FID, SSIM, LPIPS, CLIP-I and User Study. This demonstrates that the texture loss effectively strengthens the model's understanding of texture semantics by optimizing semantic alignment in the feature space, thereby enabling the generation of high-quality garments that better match the expected visual patterns.

Table 2. Ablation study on the effect of texture loss.

Methods	FID ↓	SSIM ↑	LPIPS ↓	CLIP-I ↑	User Study ↑
W/O Texture Loss	12.34	0.802	0.177	0.77	62
W/ Texture Loss	**11.55**	**0.871**	**0.123**	**0.83**	**80**

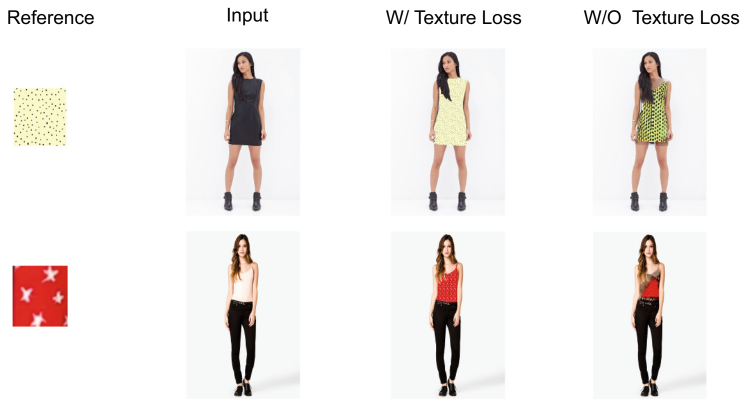

Fig. 6. Ablation study on the effect of texture loss for semantic alignment in texture synthesis.

5 Conclusion

In this paper, we presented TTEdit, a diffusion-based framework for fashion image editing that achieves high texture fidelity, accurate region localization, and strong semantic consistency. We proposed two key modules: a Mask Classification Network (MCNet) for precise editable region localization, and a Texture-Adapter for multi-scale texture injection through cross-modal attention. Comprehensive experiments demonstrate that TTEdit outperforms state-of-the-art methods in both objective metrics (FID, SSIM, LPIPS, CLIP-I) and subjective evaluations. It delivers realistic local edits, faithful texture reproduction, and robust performance across complex garment structures and materials. This work highlights the potential of diffusion models for fine-grained and controllable fashion image editing.

Acknowledgements. This work is supported by Guangxi Key R&D Program (AB25069302), the Nanning Yongjiang Talent Program (No. 2023018, No. 2019004), Guilin S&T Program (20230119-1), Graduate Innovation Project of Guangxi Academy of Artificial Intelligence and Guangxi Natural Science Foundation (No. 2025GXNS-FAA069684).

References

1. Shuai, X., Ding, H., Ma, X., et al.: A survey of multimodal-guided image editing with text-to-image diffusion models. arXiv preprint arXiv:2406.14555 (2024)
2. Zhan, F., Yu, Y., Wu, R., et al.: Multimodal image synthesis and editing: the generative AI era. IEEE Trans. Pattern Anal. Mach. Intell. **45**(12), 15098–15119 (2023)
3. Goodfellow, I., Pouget-Abadie, J., Mirza, M., et al.: Generative adversarial nets. In: Advances in Neural Information Processing Systems, vol. 27 (2014)
4. Ho, J., Jain, A., Abbeel, P.: Denoising diffusion probabilistic models. In: Advances in Neural Information Processing Systems, vol. 33, pp. 6840–6851 (2020)
5. Radford, A., Kim, J.W., Hallacy, C., et al.: Learning transferable visual models from natural language supervision. In: International Conference on Machine Learning, pp. 8748–8763. PMLR (2021)
6. Baykal, A.C., Anees, A.B., Ceylan, D., et al.: CLIP-guided StyleGAN inversion for text-driven real image editing. ACM Trans. Graph. **42**(5), 1–18 (2023)
7. Kawar, B., Zada, S., Lang, O., et al.: Imagic: text-based real image editing with diffusion models. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 6007–6017 (2023)
8. Lin, Y., Chen, Y.W., Tsai, Y.H., et al.: Text-driven image editing via learnable regions. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 7059–7068 (2024)
9. Lin, A., Zhao, N., Ning, S., et al.: FashionTex: controllable virtual try-on with text and texture. In: ACM SIGGRAPH 2023 Conference Proceedings, pp. 1–9 (2023)
10. Rombach, R., Blattmann, A., Lorenz, D., et al.: High-resolution image synthesis with latent diffusion models. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 10684–10695 (2022)
11. Ramesh, A., Dhariwal, P., Nichol, A., et al.: Hierarchical text-conditional image generation with CLIP latents. arXiv preprint arXiv:2204.06125 (2022)
12. Li, J., Li, D., Xiong, C., Hoi, S.C.H.: BLIP: bootstrapping Language-Image Pre-training for unified vision-language understanding and generation. In: Proceedings of the 39th International Conference on Machine Learning (ICML 2022), PMLR, vol. 162, pp. 12888–12900 (2022)
13. Brooks, T., Holynski, A., Efros, A.A.: InstructPix2Pix: learning to follow image editing instructions. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 18392–18402 (2023)
14. Wang, T., Ye, M.: TexFit: text-driven fashion image editing with diffusion models. In: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 38, no. 9, pp. 10198–10206 (2024)
15. Nichol, A., Dhariwal, P., Ramesh, A., et al.: Glide: towards photorealistic image generation and editing with text-guided diffusion models. arXiv preprint arXiv:2112.10741 (2021)
16. Gal, R., Alaluf, Y., Atzmon, Y., et al.: An image is worth one word: personalizing text-to-image generation using textual inversion. arXiv preprint arXiv:2208.01618 (2022)
17. Mou, C., Wang, X., Xie, L., et al.: T2I-adapter: learning adapters to dig out more controllable ability for text-to-image diffusion models. In: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 38, no. 5, pp. 4296–4304 (2024)
18. Lewis, K.M., Varadharajan, S., Kemelmacher-Shlizerman, I.: TryOnGAN: body-aware try-on via layered interpolation. ACM Trans. Graph. (TOG) **40**(4), 1–10 (2021)

19. Xie, Z., Huang, Z., Zhao, F., et al.: PASTA-GAN++: a versatile framework for high-resolution unpaired virtual try-on. arXiv preprint arXiv:2207.13475 (2022)
20. Zhang, J., Li, K., Lai, Y.K., et al.: PISE: person image synthesis and editing with decoupled GAN. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 7982–7990 (2021)
21. Raffiee, A.H., Sollami, M.: GarmentGAN: photo-realistic adversarial fashion transfer. In: 2020 25th International Conference on Pattern Recognition (ICPR), pp. 3923–3930. IEEE (2021)
22. Kirillov, A., et al.: Segment anything. In: Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), pp. 4015–4026 (2023)
23. Xiao, J., Wang, Z.: DeepLabv3+: encoder-decoder with atrous separable convolution for semantic image segmentation. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), pp. 1–5 (2019)
24. Zhao, H., Shi, J., Qi, X., Wang, X., Jia, J.: Pyramid scene parsing network. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), pp. 2881–2890 (2017)
25. Gatys, L.A., Ecker, A.S., Bethge, M.: Image style transfer using convolutional neural networks. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, pp. 2414–2423 (2016)
26. Jiang, Y., Yang, S., Qiu, H., Wu, W., Loy, C.C., Liu, Z.: Text2Human: text-driven controllable human image generation. ACM Trans. Graph. (Proceedings of SIGGRAPH) **41**(4), 1–11 (2022)
27. Yang, B., Gu, S., Zhang, B., et al.: Paint by example: exemplar-based image editing with diffusion models. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 18381–18391 (2023)
28. Chen, X., Huang, L., Liu, Y., et al.: AnyDoor: zero-shot object-level image customization. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 6593–6602 (2024)
29. Song, W., Jiang, H., Yang, Z., et al.: Insert anything: image insertion via in-context editing in DiT. arXiv preprint arXiv:2504.15009 (2025)